

Sec 5.3 (part 2)

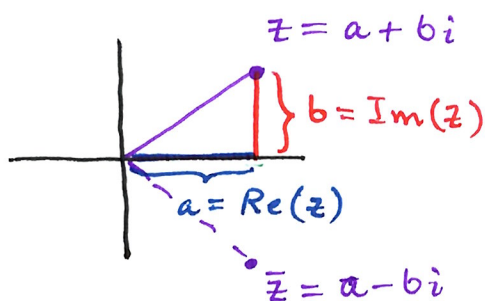
Complex numbers: $z = \underbrace{a}_{\text{Real part}} + \underbrace{bi}_{\text{Imaginary Part}}$
 $\text{Im}(z) = b$
 $\text{Re}(z) = a$

$$i^2 = -1$$

Conjugation

$$z = a + bi$$

$$\bar{z} = a - bi$$



$$z + \bar{z} = 2a = 2\text{Re}(z) \Rightarrow \text{Re}(z) = \frac{z + \bar{z}}{2}$$

$$z - \bar{z} = 2bi = 2i \cdot \text{Im}(z) \Rightarrow \text{Im}(z) = \frac{z - \bar{z}}{2i} \quad \text{Don't forget!}$$

Complex Exponential

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots$$

$$\text{try } e^{ix} = 1 + ix + \frac{i^2 x^2}{2!} + \frac{i^3 x^3}{3!} + \frac{i^4 x^4}{4!} + \dots$$

$$i = \sqrt{-1} \quad i^2 = -1, \quad i^3 = -i, \quad i^4 = 1, \quad i^5 = i, \quad i^6 = i^2 = -1, \dots$$

$$\text{so } e^{ix} = \underbrace{1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots}_{\text{(Real part)}} + i \underbrace{\left(x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots \right)}_{\text{Imag. part (Taylor Series for } \sin(x) \text{)}}$$

Euler's
Formula

$$e^{ix} = \cos x + i \sin x$$

$$\text{Re}(e^{ix}) = \cos x, \quad \text{Im}(e^{ix}) = \sin x$$

Similarly, $e^{-ix} = \cos(-x) + i \sin(-x)$

$$\boxed{e^{-ix} = \cos(x) - i \sin(x)}$$

(conjugate of e^{ix}) is $= e^{-ix}$

$$\bullet \cos(x) = \operatorname{Re}(e^{ix}) = \frac{1}{2}(e^{ix}) + \frac{1}{2}(e^{-ix})$$

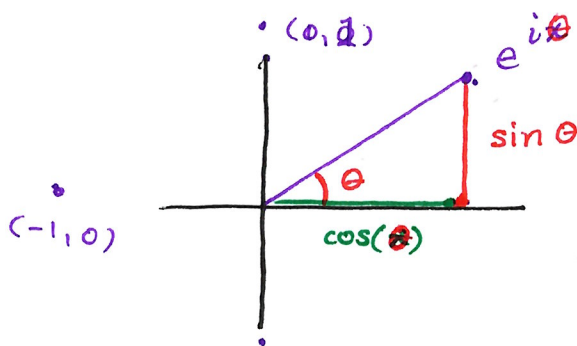
$$\bullet \sin(x) = \operatorname{Im}(e^{ix}) = \frac{1}{2i}(e^{ix}) - \frac{1}{2i}(e^{-ix})$$

$$e^{i\pi} = -1 + 0i$$

$$\cos(\pi) = -1$$

$$\sin(\pi) = 0$$

$$e^{i(\pi/2)} = 0 + 1i = i$$



$$y'' + y = 0$$

$$\text{char. eq. } r^2 + 1 = 0 \Rightarrow r^2 = -1$$

roots are $+i, -i$

$$\text{can also factor } r^2 + 1 = (r-i)(r+i)$$

$$\text{so basis solutions } \left. \begin{aligned} y_1 &= e^{ix} \\ y_2 &= e^{-ix} \end{aligned} \right\} \text{lin indep}$$

$$\text{gen solution } y = c_1 y_1 + c_2 y_2 = \boxed{c_1 e^{ix} + c_2 e^{-ix}}$$

$$\text{Hmm... } y'' + y = 0 \Rightarrow y'' = -y$$

$$y_1 = \cos x; \quad y_1'' = -\cos(x) \Rightarrow y_1 = \cos(x) \text{ is a solution}$$

$$y_2 = \sin x, \quad y_2'' = -\sin(x) \Rightarrow y_2 = \sin(x), \text{ " " " "}$$

$$\text{so every solution is } y = \tilde{c}_1 \cos(x) + \tilde{c}_2 \sin(x)$$

Prefer to use non-complex basis solutions for mechanical problems.

Ex: $y'' + 25y = 0$

char. eq. $r^2 + 25 = 0 \Rightarrow r^2 = -25 \Rightarrow r = \pm 5i$

so gen solution $y = c_1 \underline{e^{5ix}} + c_2 \underline{e^{-5ix}} \left(\{e^{ix}, e^{-ix}\} \right)$

"convert" this to $\{\cos x, \sin x\}$ basis solutions

$$y = c_1 \left(\underline{\cos(5x) + i \sin(5x)} \right) + c_2 \left(\underline{\cos(5x) - i \sin(5x)} \right)$$

$$= \underbrace{(c_1 + c_2)}_{\substack{\text{"A"} \\ \#1}} \underbrace{\cos(5x)}_{\#2} + \underbrace{i(c_1 - c_2)}_{\substack{\text{"B"} \\ \#4}} \underbrace{\sin(5x)}_{\#3}$$

gen solution can be equivalently expressed as

$$\boxed{y = A \underline{\cos(5x)} + B \underline{\sin(5x)}}$$

★ Notice the basis solutions $\{\cos(5x), \sin(5x)\}$ can be obtained by simply using $\text{Re}(e^{5ix})$, $\text{Im}(e^{5ix})$
(let's you skip conversion)

Read the examples in posted.

(incl. repeated complex roots)

Theorem: If $y(x) = a(x) + b(x)i$ (so $y(x)$ is a complex-valued function) and y is a solution of a homogeneous, linear ODE, then $a(x) = \operatorname{Re}(y(x))$ and $b(x) = \operatorname{Im}(y(x))$ are two solutions to the same ODE. (a and b may or may not be linearly indep)

Idea: Example Eq $\{y'' + 5y' - 7y = 0\}$. Because $y = a + bi$ (a, b depending on x)
 then $y = a + bi$
 $y' = a' + b'i$. Since y is a solution, then
 $y'' = a'' + b''i$

$$\hookrightarrow y'' + 5y' - 7y = 0$$

$$\Rightarrow (a'' + b''i) + 5(a' + b'i) - 7(a + bi) = 0 + 0i$$

$$\Rightarrow (\underbrace{a'' + 5a' - 7a}_0) + i(\underbrace{b'' + 5b' - 7b}_0) = \underline{0} + \underline{0}i$$

(meaning a is a solution for same ODE) (b is a solution for the same ODE)

Then, the specific form $\{y'' + 5y' - 7y = 0\}$ didn't matter; this idea would work with different #'s, or y''' , etc.

Use: Gen solution for $\{y'' + 9y = 0\}$. char poly $r^2 + 9 = 0$
 $\Rightarrow r = \pm 3i$, so $y_1 = e^{3ix}$, $y_2 = e^{-3ix}$ are two solutions. Alternatively,
 $\tilde{y}_1 = \operatorname{Re}(e^{3ix}) = \cos(3x)$
 $\tilde{y}_2 = \operatorname{Im}(e^{3ix}) = \sin(3x)$ } two other solutions, AND lin. indep!

Since eq is 2nd order, 2 lin indep solutions is enough, so
 gen solution $y = c_1 \tilde{y}_1 + c_2 \tilde{y}_2 = \boxed{c_1 \cos(3x) + c_2 \sin(3x)}$ works

Notes ★

- complex roots always come in conjugate pairs (Ex: $0 \pm 5i$, $3 \pm 2i$, $\pi \pm 7i$, ...)
- conjugate pairs of roots produce pairs of solutions:

$$\text{root pair } r = a \pm bi$$

$$\rightarrow \text{solution pair } c_1 e^{(a+bi)x} + c_2 e^{(a-bi)x}$$

$$\text{Alt: Use solution } = e^{ax} (c_1 e^{i(bx)} + c_2 e^{i(-bx)})$$

$$\begin{array}{l} e^{(a+bi)x} \text{ Real and Imag.} \\ \text{parts as basis solutions} \end{array} = e^{ax} (A \cos(bx) + B \sin(bx))$$

convert ...

$$\text{Re}(e^{(a+bi)x}) = e^{ax} \cos(bx) = \boxed{A e^{ax} \cos(bx) + B e^{ax} \sin(bx)}$$

$$\text{Im}(e^{(a+bi)x}) = e^{ax} \sin(bx) \left(\text{"a} \pm bi \text{ piece"} \right)$$

$$\text{Ex: } y^{(3)} - 3y'' + 25y' - 75y = 0$$

$$\text{ch. eqn } \underbrace{r^3 - 3r^2}_{\times 3} + \underbrace{25r - 75}_{\times 3} = 0$$

$$r^2(r-3) + 25(r-3) = 0$$

$$(r-3)(r^2+25) = 0$$

$$\begin{array}{ll} r=3 & r = \pm 5i \quad (\text{alt: } r = 0 \pm 5i) \\ \text{mult 1} & \text{mult 1} \end{array}$$

So gen solution is

$$y = (3 \text{ piece}) + (\pm 5i \text{ piece})$$

$$= c_1 e^{3x} + (c_2 \underline{\cos(5x)} + c_3 \underline{\sin(5x)})$$

(we skipped "conversion") ↗

★

for $\cos(5x)$, $\sin(5x)$ we basically took real
and imaginary parts of solution

$$e^{i(5x)} = \boxed{\cos(5x)} + i \boxed{\sin(5x)}$$

Read "order 4 example"

(Next page)

Order 4 example

Ex : $2y^{(4)} + 7y'' + 3y = 0$

ch. eq : $2r^4 + 7r^2 + 3 = 0$

(only powers of r^2 present)

trick : $s = r^2$

ch. eq $\rightarrow 2s^2 + 7s + 3 = 0$

quad form $\Rightarrow s = \frac{-7 \pm \sqrt{49 - 24}}{4}$

$$s = \frac{-7 \pm \sqrt{25}}{4} = \frac{-7 \pm 5}{4} \Rightarrow s = -3 \text{ or } s = -\frac{1}{2}$$

$$\begin{array}{l} s = -3 \rightarrow r^2 = -3 \Rightarrow r = \pm \sqrt{3} i \\ s = -\frac{1}{2} \rightarrow r^2 = -\frac{1}{2} \Rightarrow r = \pm \frac{1}{\sqrt{2}} i \end{array} \left. \vphantom{\begin{array}{l} s = -3 \\ s = -\frac{1}{2} \end{array}} \right\} \begin{array}{l} \text{two} \\ \text{conjugate} \\ \text{pairs} \end{array}$$

so general solution is

$$y = (\underline{\pm \sqrt{3} i \text{ piece}}) + (\underline{\pm \frac{1}{\sqrt{2}} i \text{ piece}})$$

$$= \left(c_1 \cos(\sqrt{3} x) + c_2 \sin(\sqrt{3} x) \right)$$

$$+ \left(c_3 \cos\left(\frac{1}{\sqrt{2}} x\right) + c_4 \sin\left(\frac{1}{\sqrt{2}} x\right) \right)$$

$\text{Re}(e^{\sqrt{3}ix}) = \cos(\sqrt{3}x)$
 $\text{Im}(e^{\sqrt{3}ix}) = \sin(\sqrt{3}x)$

$\text{Re}(e^{ix/\sqrt{2}}) = \cos(x/\sqrt{2})$
 $\text{Im}(e^{ix/\sqrt{2}}) = \sin(x/\sqrt{2})$